Uniform prior-scale bounds for structured Boolean targets

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Abstract

We bound the prior probability that a Gaussian tanh network realizes a Boolean target with output margin greater than 1/2 on every input. These target-mass bounds give learning guarantees for the posterior conditioned on that global absolute margin and exact training labels, uniformly over all prior prefactors $0 < t \le 1$. Here t multiplies the weight standard deviations, and $t = 1$ is the critical scale. For each fixed threshold depth, the sample bound is polynomial jointly in input dimension, circuit resources, and learner depth at every realizable width. Fourier variation gives a second polynomial target class at every learner depth. The proof combines a critical scalar program, an exact change of Gaussian reference, and a separate short-padding construction. The finite-width intrinsic learning and critical scalar inputs are stated explicitly. The conclusions depend on those imported analytic results; the local scale-change and padding arguments are proved here.

1 Model and conclusions

Let $X_d = \{-1, 1\}^d$, let $f : X_d \to \{-1, 1\}$, and put $H = L - 1 \ge 1$. The learner has H hidden tanh layers of common width N, augmented input $(1, x)$, and scalar linear readout. The independent Gaussian entry variances are

$$W_{ij}^1 : t^2/d, \qquad W_{ik}^j : t^2/N \ (2 \le j \le H), \qquad v_i : t^2/N^2, \qquad 0 < t \le 1.$$

The posterior conditions only on $|z(x)| > 1/2$ for every $x \in X_d$ and on agreement with the noiseless training labels. Write

$$A_f = \{w : f(x)z_w(x) > 1/2 \text{ for all } x \in X_d\}, \qquad a_{N,t}(f) = -N^{-1} \log \mu_{N,t}(A_f).$$

A realizable width means $\mu_{N,t}(A_f) > 0$. The learning criterion is: for $0 < \varepsilon, \delta < 1$, with training probability at least $1 - \delta$, the posterior assigns probability strictly greater than $2/3$ to the correct label outside a set of test-distribution mass at most ε. The input distribution is arbitrary. References below to the *original posterior* mean this conditioned finite-width law.

The event A_f is the set of parameters realizing the entire target with the required signed margin. We call its prior probability the *target mass*; $a_{N,t}(f)$ is its negative logarithm divided by width. The prior prefactor t scales every weight standard deviation. The endpoint $t = 1$ is called critical, and all smaller t give a more concentrated prior. For a target circuit with h gate stages, *padding* means the $H - h$ hidden layers remaining after the circuit has been encoded in a network prefix.

We state the two learning conclusions first, then list the imported intrinsic-learning and scalar-control inputs. Sections 3–4 show how to reuse the critical construction at other scales, with separate arguments for long and short padding. Section 5 applies the construction to Fourier expansions. The closing dependency list identifies the supporting proofs; the analytic learning statements here are not claimed to be machine verified.

Theorem 1.1 (Threshold circuits, all scales and learner depths)**.** *There is a universal constant C with the following property. Suppose f is computed by a layered threshold circuit with $h \geq 1$ gate stages ending in the output bit f, width at most C_0, normalized gate margins at least one, and coefficient magnitudes at most B. Constants may be carried as ordinary circuit gates. Put*

$$R = d + C_0 + B + h + L + 4.$$

For every $L \geq h + 1$, every realizable width, and every $0 < t \leq 1$, the stated learning criterion holds whenever

$$n \geq \frac{C}{\varepsilon}\left[R^{C(h+1)^2}(1 + \log(1/\varepsilon))^C + \log(1/\delta)\right]. \tag{1}$$

If $H - h \geq C[\log(d + 2) + h \log R]$, the exponent $C(h + 1)^2$ can be replaced by $C(h + 1)$.

Theorem 1.2 (Fourier variation at every learner depth)**.** *Let*

$$A(f) = \sum_{S \subseteq [d]} |\widehat{f}(S)|, \qquad f = \sum_S \widehat{f}(S) \prod_{i \in S} x_i.$$

There are universal C, c such that for every $L \geq 2$, every realizable width, and every $0 < t \leq 1$, it suffices that

$$n \geq \frac{C}{\varepsilon}\left[(d + L + A(f) + 2)^c(1 + \log(1/\varepsilon))^c + \log(1/\delta)\right]. \tag{2}$$

In particular, the bound is polynomial for every parity, uniformly in its degree and in the original Gaussian scale.

Neither theorem includes arbitrary polynomial-size circuits of unrestricted threshold depth with a polynomial bound independent of h. Nor does a tanh teacher become a threshold circuit merely by replacing its hidden activations by their signs.

2 Explicit analytic inputs

Set

$$b = t^{-2} - 1, \qquad \theta = b/2, \qquad c_t = \frac{t^{-2} - 1 + 2\log t}{2}.$$

Let V_* be the critical scalar Bellman baseline: the limiting minimum scalar continuation cost as its starting variance tends to zero. This input comes from *Depth and learning bounds for critical Bayesian neural networks*; the precise source locations are listed at the end. It is independent of the target and serves as the reference cost in the bounds below. Define $\mathcal{R}(D) \geq 1$ by

$$\mathcal{R}(D) - 1 - \log \mathcal{R}(D) = 2D.$$

The following two intrinsic learning inputs are proved in the companion research notes listed at the end.

Proposition 2.1 (Uniform intrinsic inputs)**.** *There are universal polynomial degrees such that the original posterior sample bound, with $a = a_{N,t}(f)$ or any deterministic upper bound for it, is polynomial in $d, L, 1 + t^2 a$ and*

$$P_t(a + t^{-2}, H) = \max\{1, t^H \mathcal{R}((a + t^{-2})/H)^{H/2}\}, \tag{3}$$

apart from the usual factor $\varepsilon^{-1}(1+\log(1/\varepsilon))^C$ *and additive* $C\varepsilon^{-1}\log(1/\delta)$*. There is also a sharper bound, polynomial in* $d, L, 1+t^2 a$ *times* $\exp(CJ)$*, where*

$$E = \big(a + 4/(Ht^2) - Hc_t - V_* + b\big)_+, \qquad J = \min\{\sqrt{HE}, E/b\}. \tag{4}$$

At $b = 0$*, the second argument is omitted. Both statements hold at every realizable width, uniformly over* $0 < t \le 1$*.*

These inputs use a bounded dual certificate with arbitrarily chosen action slack, scale-normalized importance quadrature, and a final split between large widths and the full finite-width VC class. In particular, they contain no separate inverse power of t.

The critical target inputs are as follows. A threshold prefix of depth h admits deterministic rare groups with seed fraction p_h such that, writing $S = \log(d+2) + h\log R$,

$$\log(1/p_h) \le CS, \qquad \text{critical prefix cost} \le 1/(8L). \tag{5}$$

At every $N \ge \exp(CS)$ its target score is uniformly accurate to $\exp(-CS)$, after increasing constants, and its actual target component p_N lies in $[p_h, 2p_h]$. Its diagonal is at most one. For $H - h \ge CS$, the critical continuation can be chosen with charged scalar costs D_j satisfying

$$\text{total action} \le V_* + CS^2/H, \quad \sum_j \sqrt{D_j} \le CS, \quad \sum_j D_j \le CS, \tag{6}$$

and all tracked components at least $\exp(-CS)$. The final increasing scalar path, including readout, has cost at most three. The critical finite-width channel theorem has metric amplification $\exp(CS)$ for this program. These statements are the quantitative finite-prefix, occupation, and scalar-steering inputs from the critical analysis.

The square-root estimate in (6) concerns the *charged scalar entropy upper costs*. It cannot be replaced by a smaller vector KL after data processing when bounding scalar second moments.

3 Changing scale without changing the physical scalar program

Lemma 3.1 (Exact change of reference)**.** *If* ρ *is a scalar control law with* $m = \mathbb{E}_\rho Z^2$*, then*

$$\mathrm{D}(\rho\|\mathcal{N}(0,t^2)) = \mathrm{D}(\rho\|\mathcal{N}(0,1)) + c_t + \theta(m-1). \tag{7}$$

Moreover, if $D = \mathrm{D}(\rho\|\mathcal{N}(0,1))$*, then* $(m-1)_+ \le C(\sqrt{D} + D)$*.*

Proof. Equation (7) is the Gaussian log-density ratio integrated under ρ. The entropy variational principle, applied to the square of a standard Gaussian, gives $D \ge (m - 1 - \log m)/2$; inversion for $m \ge 1$ gives the bound. □

Lemma 3.2 (Smaller residual noise preserves the target component)**.** *Put* $\psi_r(z) = \mathbb{E}\tanh(z + \sqrt{r}G)$*. For* $z > 0$*,* $\psi_r(z)$ *is nonnegative and nonincreasing in* r*. A critical Gaussian scalar stage with the same target component and no larger residual can therefore be reused at any* $t \le 1$*, while tracking exactly the critical target component and no larger residual.*

Proof. The heat identity gives $\partial_r \psi_r(z) = \frac{1}{2}\mathbb{E}\tanh''(z + \sqrt{r}G) \le 0$. Indeed $\tanh''$ is odd and negative on the positive half-line, and a Gaussian shifted by $z > 0$ puts greater density at u than at $-u$ for $u > 0$. Continuity treats $r = 0$.

For a Gaussian signal with variance av and residual variance r, the total next diagonal is $F(av + t^2 r)$, where $F(q) = \mathbb{E}\tanh^2(\sqrt{q}G)$ is increasing. The available conditional mean component increases when the residual decreases, whereas the total diagonal decreases. If a normalized score b_0 has component $v_0 \geq v$, replace it by

$$b = \sqrt{v/v_0}\, b_0 + \sqrt{1 - v/v_0}\, \xi,$$

where ξ is an independent uniform sign included under both reference and controlled laws. This keeps unit score norm and component exactly v, costs zero KL, and increases its pointwise bound by at most one. The new residual is the diagonal minus v. □

Proposition 3.3 (Uniform long-padding target mass)**.** *For the threshold target and S above, if $H - h \geq CS$, then for every $N \geq \exp(CS)$ and every $0 < t \leq 1$,*

$$a_{N,t}(f) \leq Hc_t + V_* + C(\theta S + S^2/H). \tag{8}$$

The constants and width threshold are independent of t.

Proof. Keep the critical prefix group fractions and mean shifts. Their Gaussian entropy costs are divided by t^2, so the prefix cost is at most $1/(8Lt^2)$. The Gaussian noise only decreases, preserving the prefix estimates and allowing the same seed component.

Reuse the physical critical Gaussian scalar laws, with the score downgrade from the preceding lemma. The final non-Gaussian scalar path uses the same normalized law as at criticality; its stable-lift estimate improves when residual noise decreases. Equation (7) and (6) charge at most one c_t per hidden stage and an additional $C\theta S$. The readout costs at most three at criticality, so its factor $t^{-2} = 1 + 2\theta$ adds only $C\theta$. This proves (8) for the ideal controlled path, with the coefficient of V_* exactly one.

For the finite-width lift, the physical metric multiplier of a scalar law is at most $\sqrt{\max\{t^2, m\}} \leq \sqrt{\max\{1, m\}} \leq \exp(C\sqrt{D})$. Thus total amplification is $\exp(CS)$ independently of t. Tracked score bounds and inverse components are also $\exp(CS)$. Use action slack $\tau = 1/(Ht^2)$ and controlled success loss

$$\kappa = \frac{c}{H(1 + t^2\overline{a})},$$

where $\overline{a}$ is the displayed deterministic upper budget. Since $t^2 c_t \leq 1/2$ and $t^2\theta \leq 1/2$, $t^2\overline{a} \leq C(H + S + S^2/H)$. Hence κ^{-1} is polynomial independently of t. The channel concentration and binary-KL comparison therefore require only $N \geq \exp(CS)$. Their added $1/(Ht^2) = (1 + 2\theta)/H$ is absorbed in (8). □

Applying (4) to (8) gives $J = O(S)$: if $b \geq S/H$, use E/b; otherwise use $\sqrt{HE}$. All normalized action factors and the width threshold are at most $\exp(CS)$. This proves the long-padding part of Theorem 1.1.

4 Short padding

Let $M = H - h$ and retain the same seed. Fix a universal small $p_* > 0$ and put $S_0 = \log(p_*/p_h)$; increase the resource constants so that $S_0 \geq 1$ and $S_0 \leq CS$. If $M = 0$, a direct readout has normalized action at most C/p_h.

For $1 \leq M < CS$, prescribe components growing geometrically from p_N to p_*. A Gaussian scalar variance multiplier in

$$1 \leq a_j \leq 128 \exp(S_0/M) \tag{9}$$

achieves each required increase exactly, uniformly in $t \le 1$. For completeness, if the target component is v and the next one is $w \le p_* = 1/128$, taking $a = 128w/v$ gives total preactivation variance at most two. Gaussian integration by parts and $\mathbb{E}\operatorname{sech}^2(\sqrt{u}G) \ge 1/6$ for $u \le 2$ give an available component at least $av/36 > w$; at $a = 1$ it is at most v. Continuity and the zero-cost score downgrade suffice.

The resulting finite-width budget obeys

$$t^2\overline{a} \le C\{1 + M\exp(S_0/M)\}, \qquad \Gamma \le 128^{M/2}\exp(S_0/2). \tag{10}$$

For $M = 0$, replace the first right side by C/p_h. The same normalized finite-width slack argument gives a threshold $N \ge \exp(CS)$. Since $\mathcal{R}(D) \le 4D + 2$, (3) yields

$$\log P_t(\overline{a} + t^{-2}, H) \le C(h+1)(S+1). \tag{11}$$

Indeed for $M \ge 1$ the upper bound is at most $CH + C(H/M)S_0$, and $H = h + M$, $M < CS$; the case $M = 0$ follows directly with $H = h$. Equations (10)–(11) and Proposition 2.1 prove (1) at widths beyond $\exp(CS)$. All smaller widths are bounded by that same threshold, so the VC bound for the entire finite-width tanh class gives (1) there as well. This completes Theorem 1.1.

5 Fourier seeds and consequences

The critical Fourier construction supplies, for each parity χ_S, a mixture of Gaussian mean shifts and cube sign symmetrizations with mean $c_S\chi_S$, $c_S \ge 1/[8(d+1)]$, and entropy at most

$$D_* = C(d+1)^3\log^2(d+2).$$

At variance t^2/d, keep the same physical shifts. Their entropy is at most D_*/t^2, the saturation estimate improves, and symmetry still gives an exact parity mean with the same lower bound. Mixing the signed parity laws with weights proportional to $|\widehat{f}(S)|/c_S(t)$ gives exact mean $f/Z_f(t)$, where $Z_f(t) \le 8(d+1)A(f)$, at the same entropy upper bound.

Mix this controlled law into the raw first law with probability $\vartheta = (D_*L)^{-1}$. A latent tag score gives the exact component

$$v = \frac{1}{D_*LZ_f(t)^2} \ge \frac{1}{D_*L[8(d+1)A(f)]^2}, \tag{12}$$

with prefix cost at most $1/(Lt^2)$ and constant diagonal at most one. The tag is included under both laws and adds no independent entropy penalty. Use score downgrade to track the right side of (12), and initialize the virtual critical scalar comparator with diagonal one. Bounded scores and features give a polynomial finite-width seed threshold by concentration over the cube.

Now $h = 1$ and $S = O(\log(d + L + A(f) + 2))$. The long-padding proof and the short-padding proof apply unchanged, including $H = 1$ with direct readout. In (11), $h + 1$ is an absolute constant. This proves Theorem 1.2.

Corollary 5.1 (Two useful regimes)**.** *Polynomial-size Boolean formulas and deterministic or nondeterministic branching programs of polynomial length and polynomial state width have* $\exp(O(\log^2 L))$ *sample bounds when their input dimension and resources are polynomial in learner depth L, uniformly over $0 < t \le 1$ and all realizable widths.*

Proof. Balanced formulas have logarithmic threshold depth and polynomial size. For branching programs, compose their Boolean transition matrices in a balanced tree using $(AB)_{pq} = \bigvee_r (A_{pr} \wedge B_{rq})$. This gives threshold depth $O(\log T)$ and size $O(TW^3)$ for length T and width W. Thus $S = O(\log^2 L)$ and the learner has enough padding for Proposition 3.3. □

Proof dependencies and verification scope

The scale-change identity, residual monotonicity, score downgrade, and short-padding estimates are proved above. Their learning consequences also use the intrinsic and critical scalar inputs stated in Section 2. The following proof notes and source sections supply those imported analytic arguments:

- `uniform-scale-intrinsic.md` and `uniform-sharp-intrinsic.md`: normalized all-width learning.
- `finite-width-dual-certificate.md` and `uniform-baseline-gain.md`: extraction and baseline gain.
- `nearcritical-target-reuse.md`: uniform critical-program reuse.
- Section 4 of this manuscript: the short-padding schedule and its explicit gain bound. The critical finite-width channel input below supplies its concentration and relative-entropy comparison.
- The critical-depth manuscript's finite circuit prefix, scalar steering, occupation, finite channel, and controlled Fourier seed sections: the quantitative critical inputs (5)–(6).

The named proof notes accompany this manuscript in the `proof-notes/` directory. The critical-depth manuscript is *Depth and learning bounds for critical Bayesian neural networks*, by ChatGPT-6 Astra, Kaarel Hänni, and Dmitry Vaintrob (2026), in the repository directory `works-in-progress/critical-depth/source/`. Its scalar baseline is defined in `critical-rate-foundations.tex`; the finite-width channel and Fourier seed inputs are in `critical-finite-width-channel.tex` and `critical-controlled-fourier-seeds.tex`. These analytic learning statements are not claimed to be machine verified.